Solution

$$I = \int_0^{\frac{\pi}{4}} \frac{\sin 2x\, dx}{\sin^4 x + \cos^4 x} = \int_0^{\frac{\pi}{4}} \frac{\sin 2x\, dx}{\left(\sin^2 x + \cos^2 x\right)^2 - 2\sin^2 x \cos^2 x}$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin 2x\, dx}{1 - \frac{1}{2}(2\sin x \cos x)^2} = \int_0^{\frac{\pi}{4}} \frac{2\sin 2x\, dx}{2 - \sin^2 2x} = \int_0^{\frac{\pi}{4}} \frac{2\sin 2x\, dx}{1 + \cos^2 2x}.$$

Put $u = \cos 2x$, Then, $du = -2\sin 2x\, dx$.

When $x = 0$, we have $u = \cos 0 = 1$. When $x = \frac{\pi}{4}$, we have $u = \cos\frac{\pi}{2} = 0$.

$$\therefore I = \int_1^0 \frac{-du}{1+u^2} = \int_0^1 \frac{du}{1+u^2} = \left[\tan^{-1} u\right]_0^1 = \frac{\pi}{4}.$$

Example 9.18

Prove that $\int_0^{\frac{\pi}{4}} \frac{dx}{a^2\sin^2 x + b^2\cos^2 x} = \frac{1}{ab}\tan^{-1}\left(\frac{a}{b}\right)$, where $a,\ b > 0$.

Solution

$$\text{Put } I = \int_0^{\frac{\pi}{4}} \frac{dx}{a^2\sin^2 x + b^2\cos^2 x} = \int_0^{\frac{\pi}{4}} \frac{\sec^2 x\, dx}{a^2\tan^2 x + b^2}.$$

Put $u = \tan x$. Then $du = \sec^2 x\, dx$.

When $x = 0$, we have $u = \tan 0 = 0$. When $x = \frac{\pi}{4}$, we have $u = \tan\frac{\pi}{4} = 1$.

$$\therefore I = \int_0^1 \frac{du}{a^2u^2 + b^2} = \frac{1}{a^2}\int_0^1 \frac{du}{u^2 + \left(\frac{b}{a}\right)^2} = \frac{1}{a^2}\left[\frac{a}{b}\tan^{-1}\left(\frac{au}{b}\right)\right]_0^1 = \frac{1}{ab}\tan^{-1}\left(\frac{a}{b}\right).$$

We derive some more properties of definite integrals.

Property 6

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

Proof

Let $u = a+b-x$. Then, we get $dx = -du$.

When $x = a$, $u = a+b-a = b$. When $x = b$, we get $u = a+b-b = a$.

$$\therefore \int_a^b f(x)dx = \int_b^a f(a+b-u)(-du) = \int_a^b f(a+b-u)du$$

$$= \int_a^b f(a+b-x)dx.$$

Note

Replace a by 0 and b by a in the above property we get the following property

$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$.

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